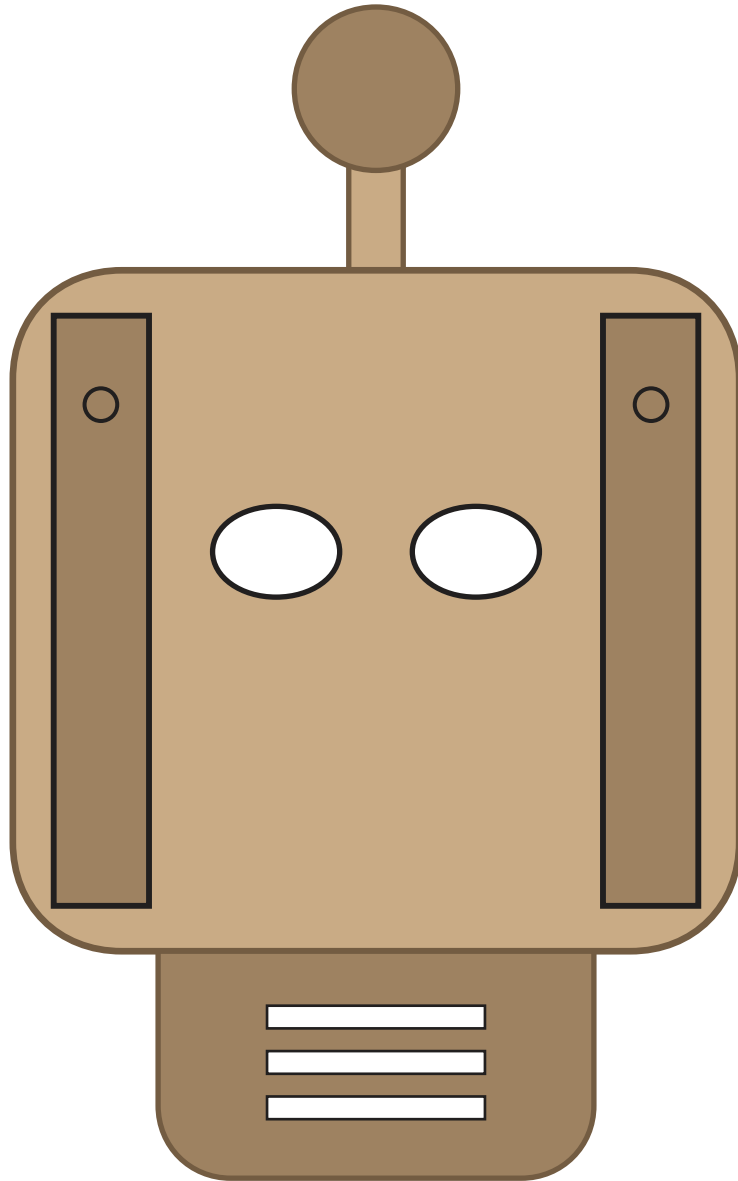




FRIENDLY ROBOT MASK

step by step guide



MEASURE TWICE • CUT ONCE • HAVE FUN

Cardboard Studio — our family workshop



What you need

MATERIALS

- Corrugated cardboard — one medium box (single-wall is fine; cereal-box card is too floppy)
- Flat elastic, about 20 inches (3/8 inch wide works great)
- White glue or hot glue
- Strong tape for the antenna tab (masking or packing tape)
- Paint, markers, bottle caps and foil to decorate (optional)

TOOLS

- Scissors or craft knife (grown-up job!)
- Metal ruler
- Pencil
- Printer + tape for the template

FINISHED SIZE

8" x 13"

DIFFICULTY

2 of 3

BUILD TIME

About 1–1.5 hours (plus
drying time)

AGES

5+ with a grown-up

SAFETY: Craft knives and hot glue are for grown-ups. Kids trace, fold, tape and decorate!



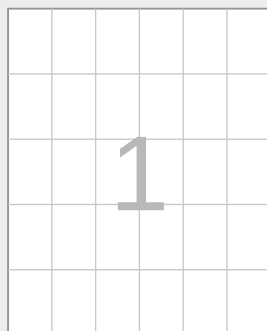
Paper size

INTERNATIONAL (EUROPEAN)

A4



210 x 297 mm
8-1/4 x 11-3/4 in



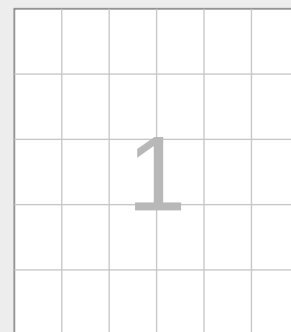
Template part has more height

NORTH AMERICA

Letter size



8.5 x 11 in
216 x 279 mm



Template part has more width



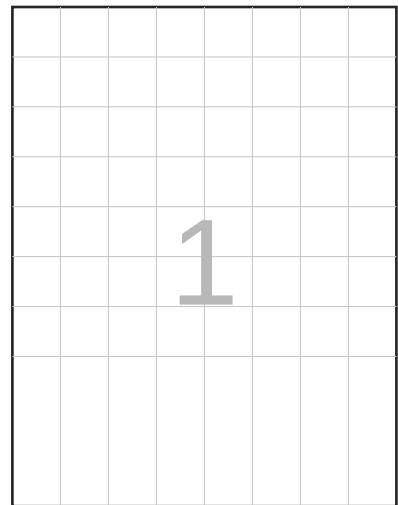
Be sure your printer
is set to print in 100%
ACTUAL SIZE



Print

Page Sizing & Handling

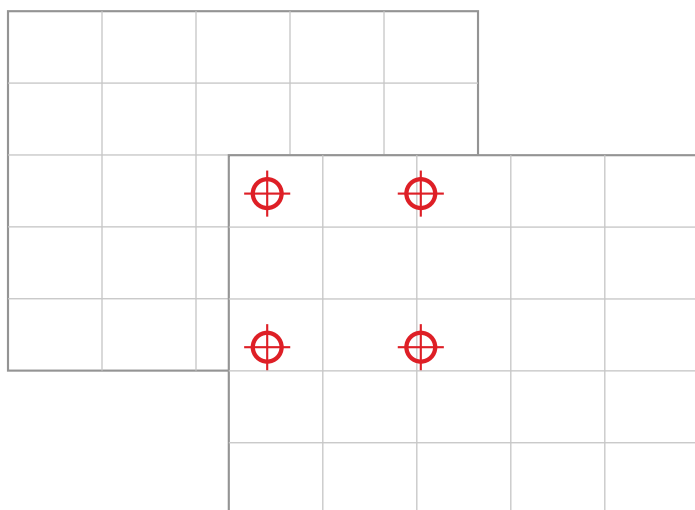
- ☐ Fit
- ☒ **Actual size** ✓
- ☐ Shrink oversized pages
- ☐ Custom Scale: 100%



Page 1 of 4

By default settings the printer shrinks the image to fit its printing area and your template will come out smaller. Each printer has its own paper margin and will not print right to the edge — that is OK: every page includes extra overlap so no detail is lost.

Pages overlap and positioning



① **Overlap**

All pages include an extra margin for overlap, so no details of the template are lost. The exact overlap depends on your printer.

② **1 inch grid (2.54 cm) / 3 inch grid (7.62 cm)**

The grid helps you align the pages perfectly in place. Light lines are 1 inch apart; heavy lines are 3 inches apart.

③ **3x3 inch square**

Overlap the pages until the visible squares measure exactly 1 inch and the heavy lines form 3x3 inch squares — your pattern will automatically align. Tape the pages together.

Your template pages arrange like this:



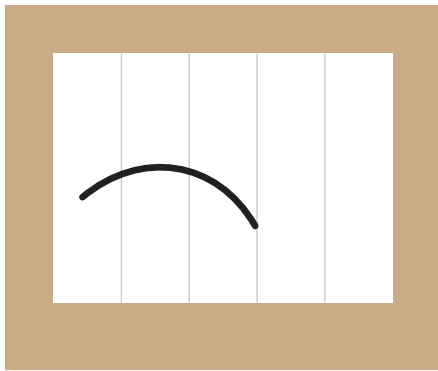
LINE UP THE GRID — THEN TAPE THE PAGES

Tracing

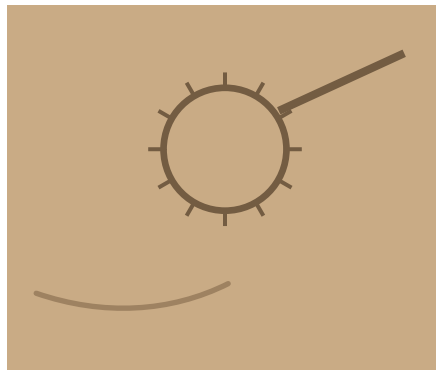
Method 1.

1. Cut Out the Template: carefully cut out the shapes from the printed template using scissors.
2. Secure the Template: place the template pieces on the cardboard and secure with tape or small weights so nothing slips.
3. Trace the Outline: use a pencil or pen to trace around the edges of the template onto the cardboard.
4. Check Your Lines: double-check every piece is traced before removing the template.

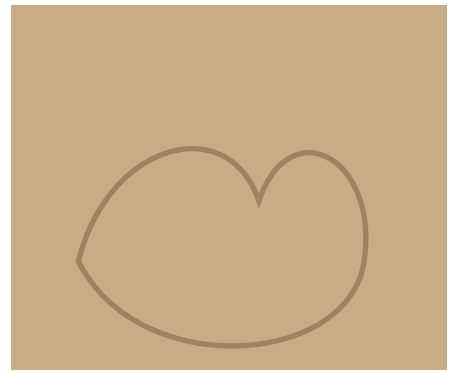
Method 2.



- ① Tape it
and trace it



- ② Tracing tool
a pen with pressure, or a tracing
wheel



- ③ Ready to cut out

NOTE - mind cardboard structure direction

The arrow on each template part shows the direction of the corrugated cardboard flutes. Keep the arrow parallel to the little waves inside your cardboard — parts stay strong and bend where they should.

Tip: the best way to bend cardboard smoothly is to cut slits halfway through on one side. Crease the slits with your fingernail to bend the other way and keep the front smooth.





NOTE - tips and tricks



HOW TO CUT CARDBOARD

PERFECT STRAIGHT CUT

Use a sharp blade and a metal ruler. Cut in 2–3 light passes instead of one hard press.

CUT A PAPER BLEED

Measure your cardboard thickness, cut half through, then peel off the corrugated layer leaving one paper face. Glue sticks much better on the bleed.

CUT A PERFECT CIRCLE

Mark the radius on a cardboard scrap, poke a pin through one end and a pencil through the other — instant compass. Cut just outside the line, then trim.



HOW TO GLUE CARDBOARD

QUICK CONNECTION — HOT GLUE

Fix the panels with tape first, open the joint like a book, run hot glue in the groove, close and tape again. Quick and sturdy for small parts.

BIG PANELS — WHITE GLUE

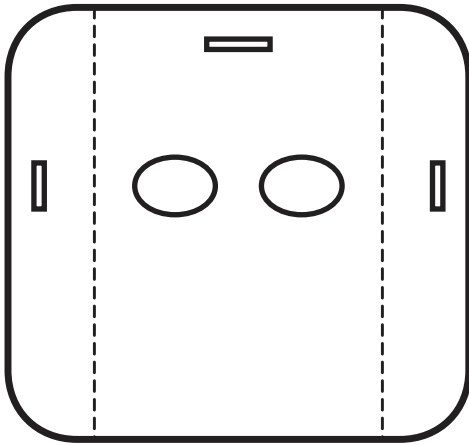
Add a little water to white glue and mix. Spread on the bottom part, lay the top parts like a mosaic so they overlap seams, press under books and let dry completely.

SEAMLESS OVERLAP

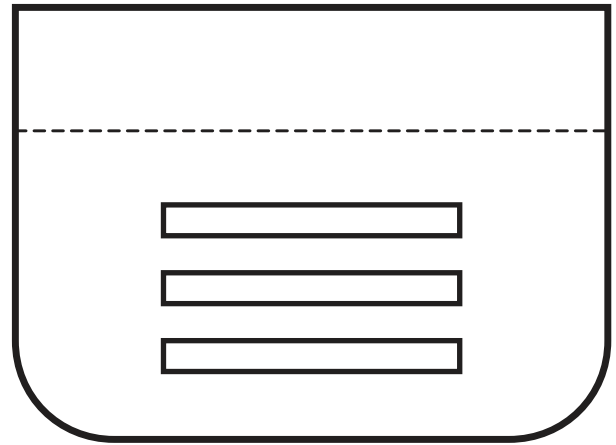
Make a paper bleed on the edges, apply white glue on the bottom part and the bleed, connect the matching parts, press and let dry.



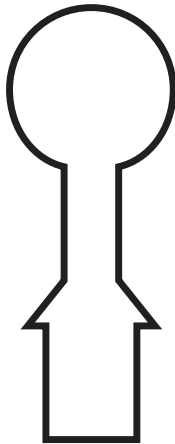
Step 1. Print, tape and trace



- ① Face Plate — trace ONCE. The ovals, small rectangles and slot are holes to cut; dashed lines are scores.



- ② Jaw Flap — trace ONCE. The dashed line is the hinge.



- ③ Antenna — trace ONCE. The narrow tab at the bottom goes through the mask.

1. Print the template at 100% actual size and tape the pages together using the grid.
2. Trace the Face Plate, Jaw Flap and Antenna once each onto cardboard.
3. Keep each corrugation arrow lined up with the little waves inside the cardboard — that keeps the mask stiff and the folds crisp.

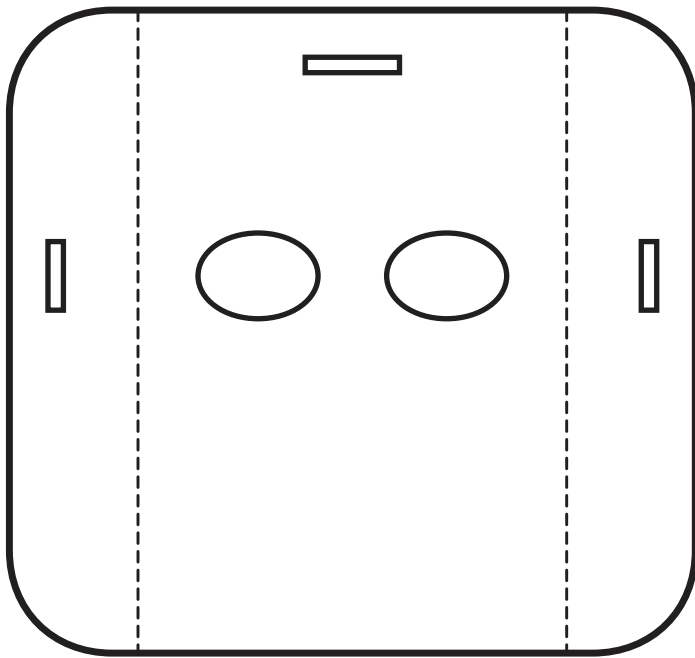
CHECK: You should have 3 pencil outlines, with the eye holes, slots and dashed lines marked inside them.

TIP: Mark the holes by poking a pencil through the template at each corner, then connect the dots.

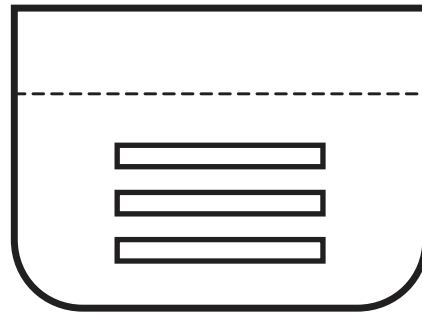


Step 2. Cut out the parts

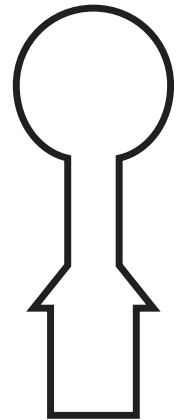
Face Plate ×1



Jaw Flap ×1



Antenna ×1



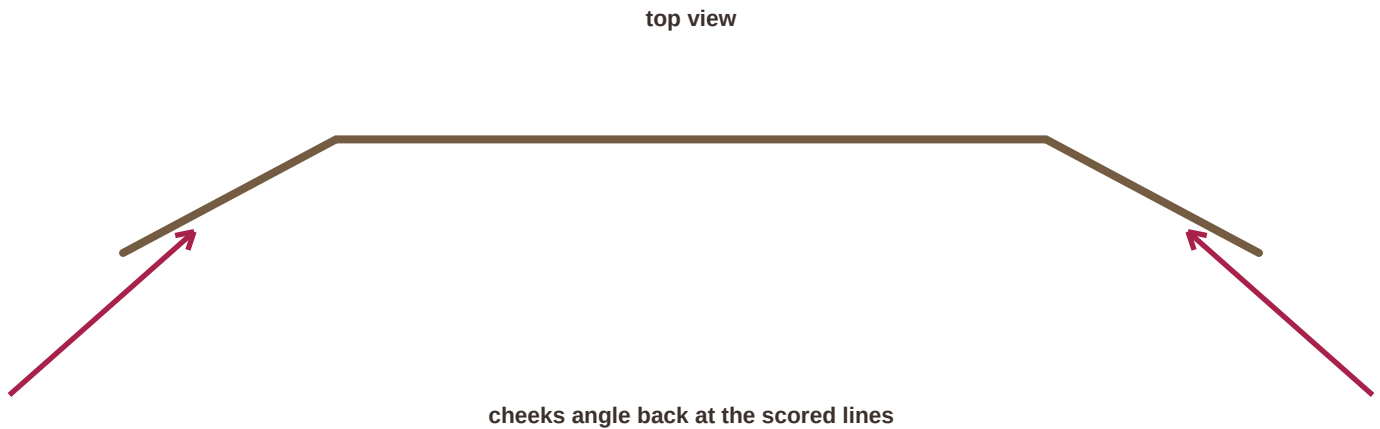
① Your full cut list — 3 pieces. Dashed lines are NOT cuts: score them half-deep only.

1. Grown-up job: cut all the solid outlines with a sharp knife and metal ruler, in 2–3 light passes.
2. Cut out the eye holes, the antenna slot, the two strap slots and the three mouth-grille slots — poke the knife tip through the middle of each and work outward.
3. Do NOT cut the dashed lines — score them lightly (half-deep) so they fold cleanly.

CHECK: 3 cut pieces; you can see through both eye holes; every dashed line is still uncut on the front.



Step 3. Curve the face plate



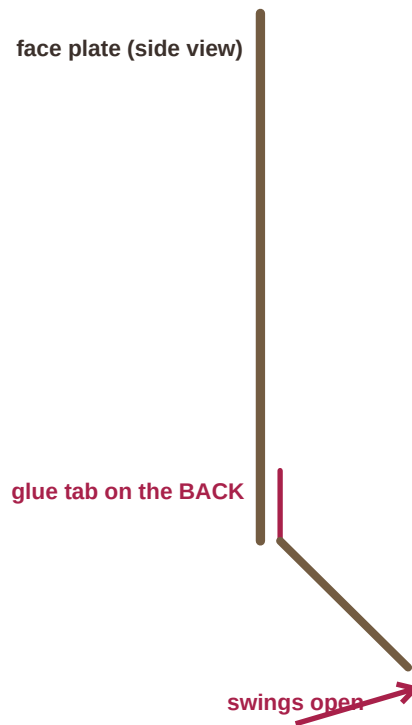
① Seen from above: crease both scored cheek lines so the sides sweep back and the mask hugs the face.

1. Lay the mask face-down and crease along each vertical scored line, folding both cheek panels gently backward.
2. Hold it up to your robot's face: eyes should line up with the eye holes and the cheeks should curve toward the ears.
3. If a fold fights you, have a grown-up deepen the score a little — never cut all the way through.

CHECK: The mask stands up like a shallow U when you set it on the table.



Step 4. Hinge the jaw



① Side view — glue **ONLY** the short tab above the scored line to the back of the face plate; the jaw below the score swings free like a drawbridge.

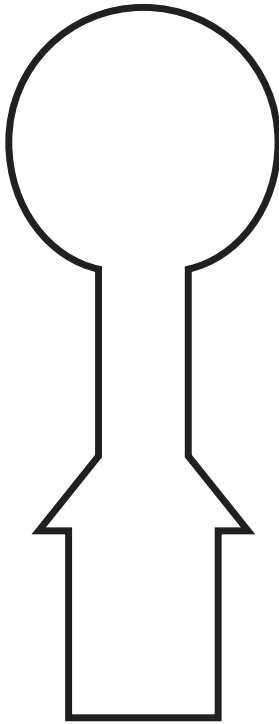
1. Fold the Jaw Flap backward along its scored hinge line so the short 1-inch tab points up.
2. Spread glue **ONLY** on the tab, then press it to the **BACK** of the face plate, centered, with the scored hinge sitting right at the face plate's bottom edge.
3. Hold for a minute, then flick the jaw gently — the grille mouth should swing open and shut.

CHECK: The jaw hangs closed on its own and opens when you flick it — and no glue touched the part below the score.

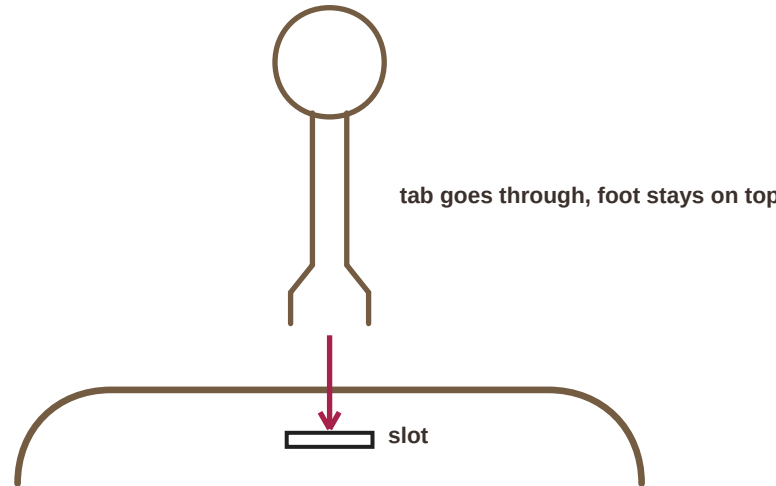
TIP: Clothespins along the bottom edge make perfect tiny clamps while the glue dries.



Step 5. Raise the antenna



① Antenna — the wide foot stops at the slot; only the narrow tab goes through.



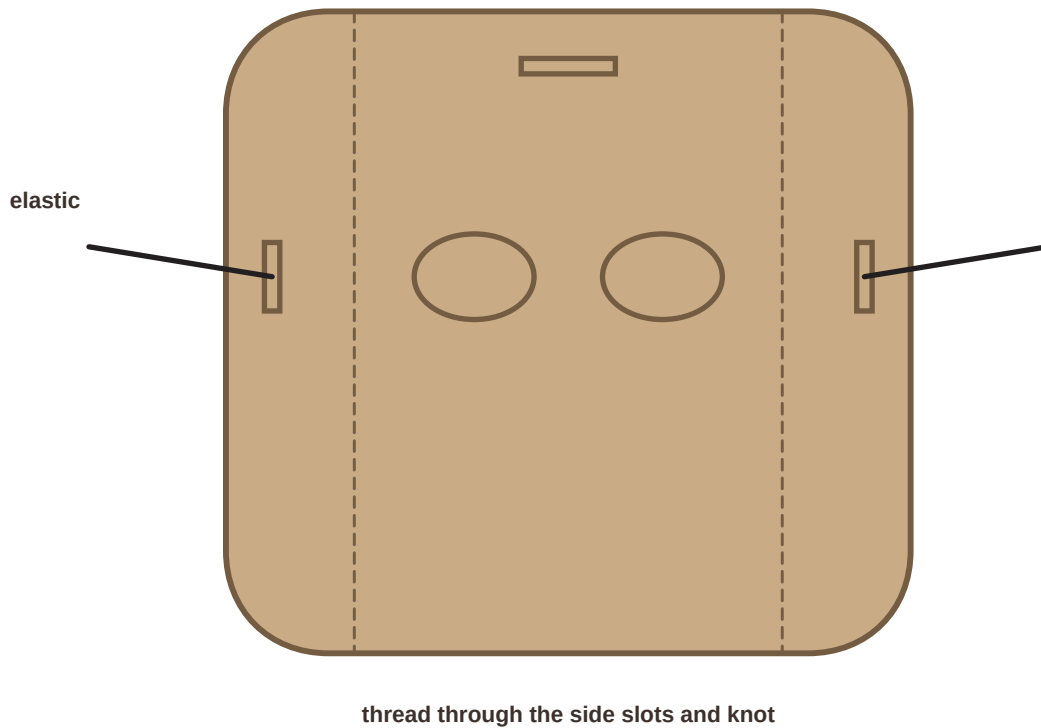
② Drop the tab through the slot from the front; the wide foot stops on the mask, and the tail gets taped flat on the back.

1. Slide the antenna's narrow tab through the slot near the top of the mask, from the front, until the wide foot sits flat.
2. On the back, fold the poking-through tab flat against the cardboard and tape it down well.
3. Give the bulb a gentle wiggle — the foot should keep it standing tall.

CHECK: The antenna stands straight up and springs back when you boop the bulb.



Step 6. Strap up and decorate

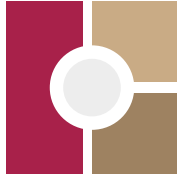


① The elastic runs from slot to slot around the back of the head — knot each end onto itself so it can't slip back through.

1. Cut about 20 inches of flat elastic, thread one end through each side slot, and tie each end back onto itself with a chunky double knot.
2. Try it on your robot and slide the knots to snug it up — comfy, not tight — BEFORE trimming the extra elastic.
3. Decorate: paint it silver or gray, add bottle-cap dials and foil rivets, and color the grille black.
4. Keep decorations clear of the eye holes so your robot can see!

CHECK: The mask stays on when your robot shakes their head, and both eyes see daylight.

TIP: No elastic? Tie on two ribbon ties instead and bow them at the back.



CARDBOARD STUDIO

MADE AT HOME, FOR HOME.

This template was designed in our family workshop. Print it, build it, smash it, rebuild it better. Share it with friends — and send us a photo of what you made!

BUILT WITH

Cardboard Studio

Design: Friendly Robot Mask • 2026